# Supplementary Material for A Self-Calibrating Single-Camera Pressure Controller for Heterogeneous Urban Traffic

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***Abstract*—This supplement holds the parameter table and the per-experiment figures and tables that the main text summarises in prose. Experiment numbers, symbols and macros are those of the main text, and every number is generated from the same raw result files by the same scripts.**

## I. Controller parameters

TABLE S1
CONTROLLER PARAMETERS. THE SINGLE-JUNCTION COLUMN IS USED IN EXPERIMENTS 1, 3, 6, 7, 8 AND 10, WITH $g_{\max} = 30$ S IN EXPERIMENTS 1 AND 3 AND 45 S IN THE OTHERS; THE CORRIDOR COLUMN IS THE CONFIGURATION OF THE SYSTEM'S NETWORK DEMONSTRATION, USED IN EXPERIMENTS 2 AND 9. EQUATION NUMBERS REFER TO THE MAIN TEXT.

| Symbol | Meaning | Single | Corridor |
|---|---|---|---|
| $g_{\min}$ | minimum green (s) | 5.0 | 5.0 |
| $g_{\max}$ | maximum green (s) | 30 / 45 | 34.0 |
| $C_{\max}$ | cycle clamp for the green floor (s) | 120 | 120 |
| $\tau$ | estimator time constant (s) | 60.0 | 60.0 |
| $y$ | yellow (s) | 3.0 | 2.0 |
| $\rho$ | all-red clearance (s) | 1.5 | 2.5 |
| $\gamma_t$ | gap-out time (s) | 2.5 | 2.5 |
| $R$ | vehicle maximum wait (s) | 60.0 | 26.0 |
| $V$ | pedestrian maximum wait (s) | 60.0 | 60.0 |
| $W$ | walk-hold allowance (s) | 12.0 | 12.0 |
| $\Theta$ | preemption cap (s) | 60.0 | 25.0 |
| $\theta$ | emergency latch (s) | 2.5 | 2.5 |
| $h$ | hysteresis margin (PCU) | 1.0 | 1.0 |
| $w_{\text{ped}}$ | pedestrian weight | 0.3 | 0.3 |
| $\gamma$ | downstream coupling | n/a | 0.4 |

## II. Single junction

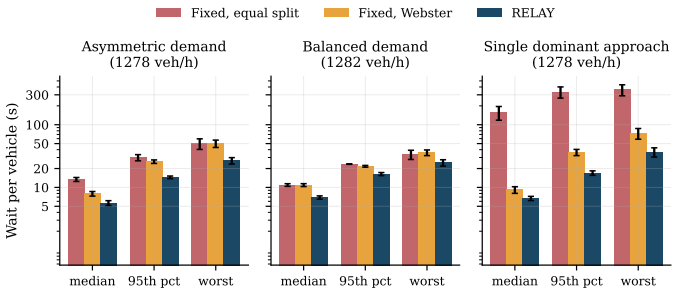



Fig. S1. Experiment 1. Median, 95th-percentile and worst observed wait per vehicle at demand multiplier 1.0, logarithmic axis. The tail is where the anti-starvation bound shows.

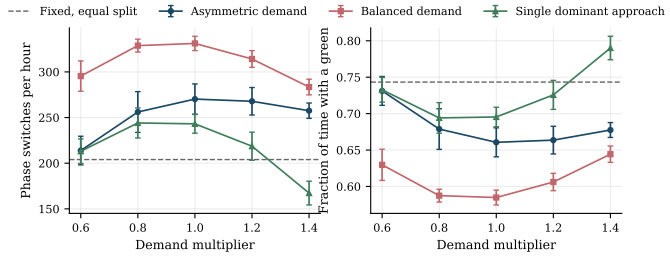



Fig. S2. Experiment 1b. Left: phase changes per hour. Right: fraction of wall time with a green displayed; the complement is clearance. Dashed: the equal-split plan, whose behaviour is independent of demand.

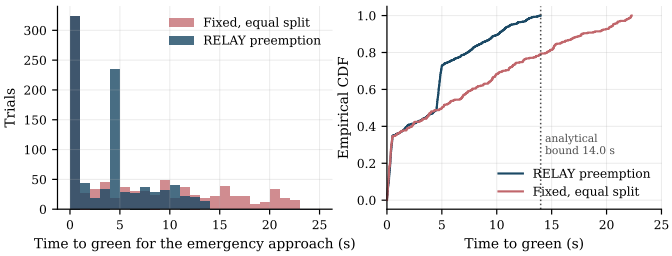



Fig. S3. Experiment 3. Left: time from first emergency detection to green, 900 trials. Right: empirical cumulative distributions against the closed-form bound of 14.0 s.

## III. Point-queue corridor

## IV. Preemption latency

## V. Perception

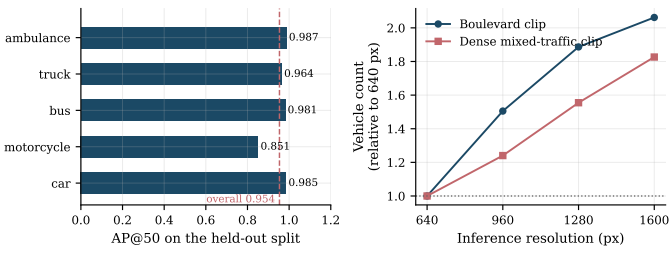



Fig. S4. Left: Experiment 4, per-class AP at IoU 0.5. Right: Experiment 5, sampled count against inference resolution, normalised to 640 px. Neither curve has flattened by 1600 px.

TABLE S2

EXPERIMENT 1. MEAN DELAY PER VEHICLE (S, STANDARD DEVIATION ACROSS 30 SEEDS IN PARENTHESES) AND THE PAIRED PER-SEED REDUCTION ACHIEVED BY RELAY AGAINST EACH FIXED-TIME BASELINE. ALL DIFFERENCES ARE SIGNIFICANT AT $p < 0.05$ BY THE WILCOXON SIGNED-RANK TEST (LARGEST $p = 0$).

| | Demand | Mean delay per vehicle (s) | | | Reduction (%) | |
|---|---|---|---|---|---|---|
| Profile | (veh/h) | Equal | Webster | RELAY | vs equal | vs Webster |
| Asymmetric | 767 | 11.0 (0.6) | 8.4 (0.6) | 5.0 (0.3) | +54.6 (3.8) | +40.1 (5.0) |
| | 1022 | 11.8 (0.7) | 8.9 (0.6) | 5.5 (0.3) | +53.0 (4.1) | +37.7 (5.0) |
| | 1278 | 13.9 (0.8) | 9.6 (0.4) | 6.4 (0.3) | +54.0 (4.5) | +33.3 (4.4) |
| | 1534 | 19.2 (3.1) | 10.7 (0.8) | 7.2 (0.4) | +61.7 (6.2) | +32.5 (4.7) |
| | 1789 | 45.1 (17.2) | 11.8 (1.0) | 8.1 (0.4) | +79.7 (6.8) | +31.3 (3.8) |
| Balanced | 769 | 10.5 (0.6) | 9.9 (0.4) | 6.1 (0.3) | +42.0 (3.3) | +38.5 (3.0) |
| | 1025 | 10.9 (0.4) | 10.5 (0.4) | 6.8 (0.3) | +38.0 (3.9) | +35.9 (3.4) |
| | 1282 | 11.4 (0.3) | 11.2 (0.4) | 7.6 (0.3) | +33.4 (3.1) | +32.0 (2.9) |
| | 1538 | 12.1 (0.6) | 11.9 (0.6) | 8.5 (0.4) | +29.7 (4.0) | +28.2 (3.7) |
| | 1794 | 13.2 (0.6) | 13.5 (0.6) | 9.6 (0.4) | +27.4 (2.8) | +28.7 (3.1) |
| Single | 767 | 14.4 (2.2) | 8.8 (0.5) | 5.4 (0.4) | +62.5 (4.0) | +39.3 (5.2) |
| | 1022 | 38.3 (16.6) | 9.8 (0.5) | 6.5 (0.4) | +79.9 (8.2) | +34.0 (5.8) |
| | 1278 | 151.8 (30.9) | 11.8 (0.9) | 7.7 (0.4) | +94.8 (1.0) | +34.7 (3.9) |
| | 1534 | 239.8 (28.3) | 17.1 (3.0) | 9.4 (0.8) | +96.0 (0.5) | +43.8 (8.7) |
| | 1789 | 307.2 (23.7) | 21.3 (4.1) | 12.8 (1.7) | +95.8 (0.5) | +38.7 (9.9) |

TABLE S3

EXPERIMENT 2. THREE-JUNCTION CORRIDOR, MEAN OVER 20 SEEDS (SD IN PARENTHESES). DELAY IS TOTAL QUEUEING DELAY PER VEHICLE ACROSS THE JUNCTIONS TRAVERSED; THROUGH TRAVEL TIME IS FOR THE TWO ARTERIAL THROUGH ROUTES (FREE-FLOW 21.6 S); EMERGENCY TRAVERSAL IS THE SINGLE EMERGENCY VEHICLE PER RUN.

| Demand (veh/h) | Policy | Mean delay per vehicle (s) | Through travel (s) | Stops per vehicle | Emergency traversal (s) |
|---|---|---|---|---|---|
| 769 | Webster, independent | 26.7 (0.9) | 39.7 (0.9) | 1.95 | 36.0 (0.0) |
| | Webster, coordinated | 25.7 (0.7) | 37.0 (0.7) | 1.78 | 24.5 (0.0) |
| | RELAY, no coupling | 11.3 (0.5) | 24.9 (0.5) | 1.09 | 22.3 (3.3) |
| | RELAY, coupled | 11.2 (0.5) | 24.9 (0.5) | 1.09 | 22.3 (3.3) |
| 1282 | Webster, independent | 30.1 (0.8) | 41.5 (0.7) | 2.09 | 53.6 (7.5) |
| | Webster, coordinated | 29.2 (1.1) | 38.0 (0.8) | 1.80 | 31.5 (0.0) |
| | RELAY, no coupling | 15.2 (0.7) | 28.3 (0.6) | 1.53 | 23.7 (4.7) |
| | RELAY, coupled | 15.2 (0.6) | 28.5 (0.6) | 1.53 | 23.5 (4.7) |
| 1794 | Webster, independent | 36.7 (3.3) | 43.1 (1.3) | 2.10 | 30.1 (6.5) |
| | Webster, coordinated | 35.3 (3.6) | 39.3 (0.9) | 2.03 | 24.0 (0.0) |
| | RELAY, no coupling | 20.2 (0.8) | 32.1 (0.7) | 1.87 | 27.6 (8.5) |
| | RELAY, coupled | 20.2 (0.8) | 32.3 (0.7) | 1.87 | 26.7 (8.3) |

TABLE S4

EXPERIMENT 3. DELAY FROM FIRST EMERGENCY DETECTION TO GREEN ON THE EMERGENCY APPROACH OVER 900 TRIALS. THE EQUAL-SPLIT BASELINE HAS NO PREEMPTION PATH; ITS LATENCY IS SET BY CYCLE POSITION AND IS POOLED.

| Demand | Policy | Mean | SD | Median | 95th pct | Worst |
|---|---|---|---|---|---|---|
| Light ($m$=0.6) | RELAY | 4.02 | 3.56 | 4.59 | 10.80 | 13.61 |
| Medium ($m$=1.0) | RELAY | 4.04 | 3.88 | 4.27 | 11.83 | 13.91 |
| Heavy ($m$=1.4) | RELAY | 4.16 | 3.86 | 4.53 | 11.85 | 13.99 |
| All | RELAY | 4.07 | 3.76 | 4.53 | 11.33 | 13.99 |
| All | Equal split | 7.00 | 7.04 | 5.02 | 20.59 | 22.29 |

## VI. BIASED-COUNT GRID

## VII. SUMO SINGLE JUNCTION

## VIII. SUMO CORRIDOR

## IX. ADAPTIVE BASELINES

## X. SENSITIVITY

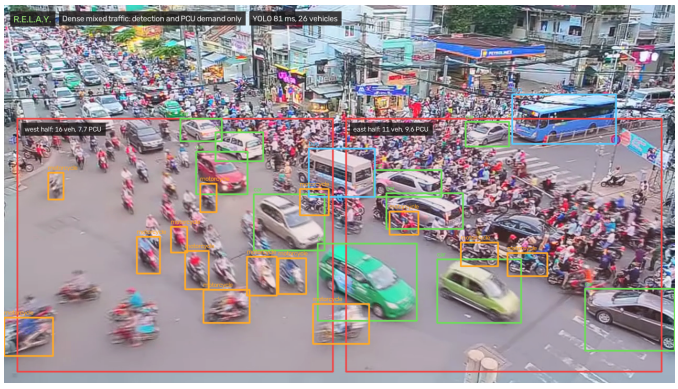



Fig. S5. Experiment 5. The stack on dense motorcycle-dominated footage. The failure mode is visible: a few dozen vehicles are detected in a scene containing several hundred, so the reported PCU demand is a severe underestimate. This is the error Experiment 6b models.

TABLE S5
EXPERIMENT 6B. THE 72-CELL GRID UNDER BIASED COUNT MODELS APPLIED TO WHAT THE CONTROLLER SEES. WON COUNTS CELLS WITH A POSITIVE REDUCTION AGAINST BOTH FIXED PLANS; WORST AND MEAN ARE THE PAIRED REDUCTION AGAINST WEBSTER OVER THE GRID.

| Count model | Won | Worst (%) | Mean (%) |
|---|---|---|---|
| Saturating, $q_{1/2} = 8$ | 71/72 | $-8.9$ | $+36.0$ |
| Saturating, $q_{1/2} = 4$ | 71/72 | $-22.1$ | $+35.3$ |
| Scale 0.8 | 71/72 | $-0.0$ | $+36.6$ |
| Scale 0.6 | 72/72 | $+0.9$ | $+36.6$ |
| Scale 0.4 (lone vehicle rounds to 0) | 51/72 | $-36.6$ | $+3.9$ |
| Scale 0.4, presence floor 1 | 72/72 | $+0.4$ | $+36.4$ |
| Scale 0.8, lone vehicle reads 0 | 50/72 | $-36.6$ | $+3.8$ |
| Scale 0.6, lone vehicle reads 0 | 50/72 | $-36.6$ | $+3.9$ |
| Scale 0.8, presence floor 1 | 71/72 | $-0.0$ | $+36.6$ |
| Scale 0.6, presence floor 1 | 72/72 | $+0.9$ | $+36.6$ |

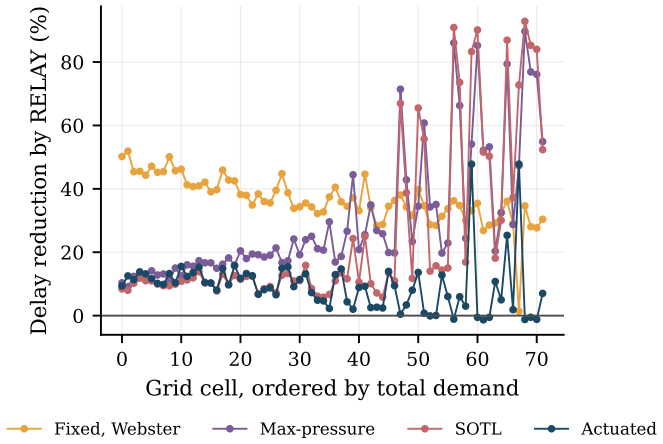



Fig. S6. Experiment 7. Paired delay reduction of RELAY against each baseline over the 72 cells, ordered by demand. Positive favours RELAY.

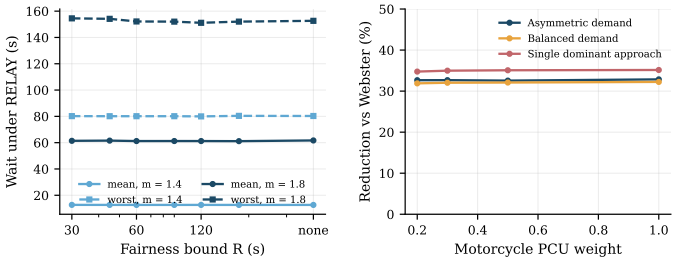



Fig. S7. Experiment 10. Left: mean and worst wait against the fairness bound $R$ on the single-dominant shape at 1789 and 2300 veh/h. Right: reduction against Webster as a function of the motorcycle PCU weight with a two-wheeler share of 80% on one axis and 30% on the other.

TABLE S6
EXPERIMENT 8. SUMO, MEAN TIME LOSS PER VEHICLE (S, SD OVER 10 SEEDS) AND PAIRED REDUCTION OF RELAY AGAINST EACH BASELINE.

| | Demand | Mean time loss (s) | | | | Reduction (%) | | |
|---|---|---|---|---|---|---|---|---|
| Profile | (veh/h) | Webster | Max-pr. | Actuated | RELAY | vs Web. | vs M-p. | vs Act. |
| Asymmetric | 767 | 7.6 (0.4) | 5.2 (0.4) | 7.7 (0.4) | 3.5 (0.2) | +53.0 | +31.8 | +53.7 |
| | 1278 | 8.2 (0.5) | 7.0 (0.4) | 8.4 (0.4) | 4.8 (0.3) | +40.9 | +30.5 | +42.6 |
| | 1789 | 8.7 (0.4) | 8.3 (0.5) | 9.0 (0.6) | 6.5 (0.5) | +25.5 | +21.8 | +28.4 |
| Balanced | 769 | 9.2 (0.3) | 6.4 (0.4) | 7.8 (0.3) | 4.1 (0.3) | +55.5 | +36.1 | +47.2 |
| | 1282 | 9.7 (0.4) | 8.4 (0.3) | 8.7 (0.3) | 5.8 (0.3) | +40.1 | +30.8 | +33.0 |
| | 1794 | 10.5 (0.4) | 10.0 (0.9) | 9.5 (0.6) | 7.8 (0.4) | +25.8 | +21.5 | +18.1 |
| Single | 767 | 7.7 (0.4) | 5.1 (0.5) | 8.1 (0.3) | 3.4 (0.2) | +55.5 | +32.4 | +57.4 |
| | 1278 | 8.4 (0.4) | 7.1 (0.5) | 8.8 (0.4) | 5.1 (0.3) | +39.1 | +28.4 | +42.6 |
| | 1789 | 8.9 (0.3) | 8.8 (0.5) | 9.6 (0.5) | 6.9 (0.5) | +22.9 | +22.3 | +28.5 |

TABLE S7
EXPERIMENT 9. SUMO CORRIDOR WITH 150 M LINKS, MEAN TIME LOSS PER VEHICLE (S, SD OVER 10 SEEDS) AND ARTERIAL THROUGH TRAVEL TIME (S).

| Demand (veh/h) | Policy | Mean time loss (s) | Through travel (s) |
|---|---|---|---|
| 769 | Webster, independent | 22.2 (1.0) | 93.3 |
| | Webster, coordinated | 20.8 (0.7) | 91.0 |
| | RELAY, no coupling | 6.9 (0.5) | 78.7 |
| | RELAY, coupled | 6.7 (0.5) | 78.5 |
| 1282 | Webster, independent | 23.3 (0.4) | 94.3 |
| | Webster, coordinated | 22.6 (0.8) | 92.7 |
| | RELAY, no coupling | 10.9 (0.6) | 79.9 |
| | RELAY, coupled | 10.8 (0.5) | 80.0 |
| 1794 | Webster, independent | 26.6 (1.2) | 96.0 |
| | Webster, coordinated | 26.0 (2.0) | 93.5 |
| | RELAY, no coupling | 17.2 (1.8) | 82.3 |
| | RELAY, coupled | 17.9 (1.7) | 82.4 |